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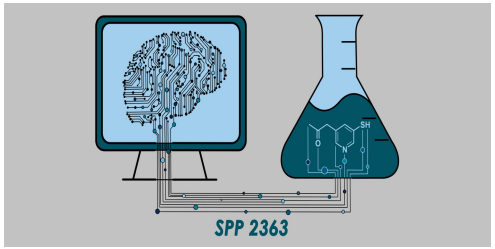


Reinforcement Learning in the Context of Molecular Design - Fundamentals and Applications

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University of Hamburg, ZBH - Center for Bioinformatics, Albert-Einstein-Ring 8-10, Hamburg, Germany

05.11.2026 - SPP PhD Tutorial - Monthly Meeting



Agenda

1 **Hook & Motivation:** The Optimization Spectrum

2 **Foundation:** Reinforcement Learning, Thoughts Through Molecule Design

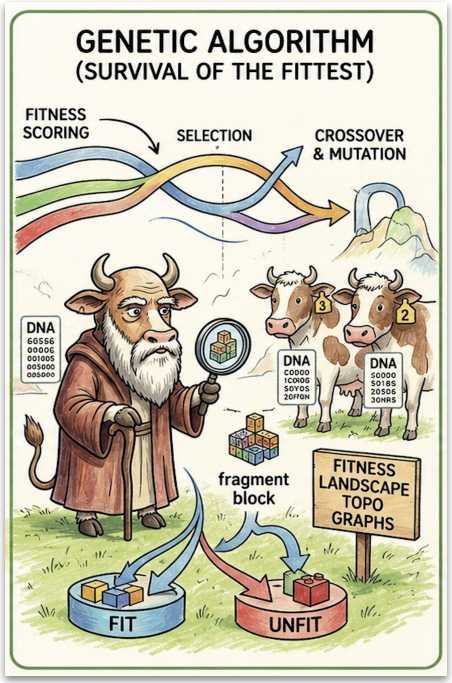
3 **Landscape:** How Reinforcement Learning is used in Molecule Design, Literatures by Taxonomy

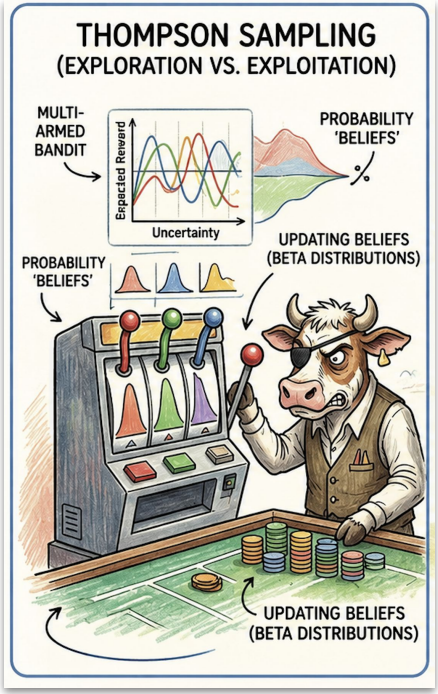
4 **Case Study:** Goal-Conditioned Reinforcement Learning over Combinatorial Fragment Spaces

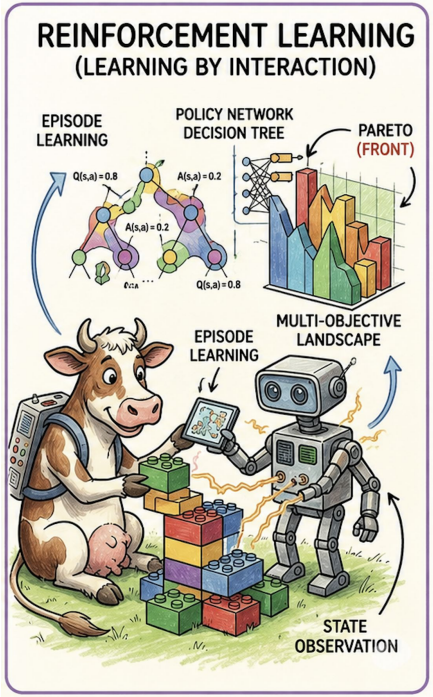
5 **References**

1 Hook & Motivation - The Optimization Spectrum

Heuristics vs. Bayesian Probabilities vs. Markovian Policies







Gemini - Nano Banana

2.1 Foundation - Three Learning Paradigms

Navigating the middle ground between supervised imitation and unsupervised discovery through evaluative feedback

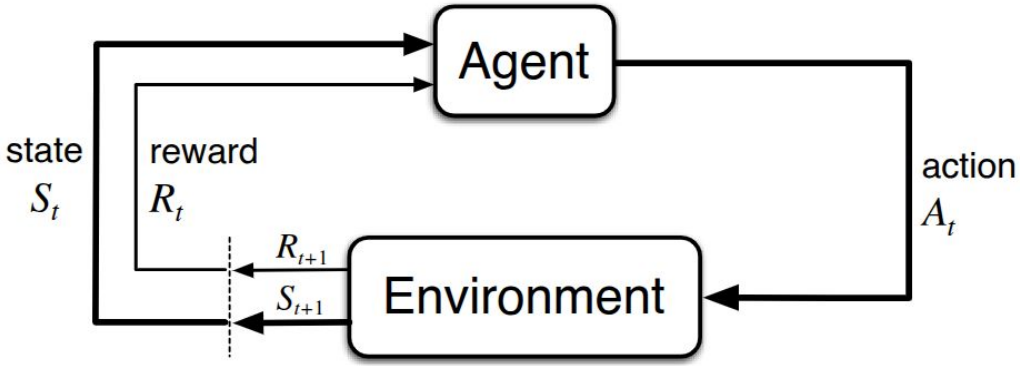
	Supervised	Unsupervised	Reinforcement
Data	Labeled pairs *(x, y)*	Unlabeled x	Self-generated via interaction
Feedback	Correct answer	None	Scalar reward
Goal	Predict *y*	Discover Structure	Maximize cumulative reward
Example	Property prediction	Clustering	Game robotics, molecule design

- **Supervised:** instructive feedback
- **RL:** evaluative feedback

2.2 Foundation - The Agent-Environment Loop

Agent observes states, executes policy-driven actions and shapes its trajectory through environmental feedback

- **Agent**: the learner and decision-maker
- **Environment**: everything outside the agent
- **State** S_t: the current situation
- **Action** A_t: the agent's decision
- **Reward** R_{t+1} , **next state** S_{t+1}: the environment's response
- **Policy** $\pi(a|s)$**:** the mapping from state to action
- **Trajectory**: $S_0, A_0, R_1, S_1, A_1, R_2$



Sutton & Barto, Reinforcement Learning: An Introduction, 2nd ed., MIT Press, 2018, Fig. 3.1.

Goal = maximize cumulative reward, not immediate reward

2.3 Foundation - What Makes RL Hard?

Hurdles: delayed rewards, the exploration-exploitation dilemma, non-stationary data distributions and black-box objectives

1. **Credit assignment:** reward delayed and sparse
   → Which of 60 moves won the chess game?
   → Which fragment made the molecule active?
2. **Exploration vs. exploitation:**
   →Trade-off between known-good and unknown-maybe-better
   →Bandit: keep pulling the best-known arm, or try a new one?
3. **Non-i.i.d. data**: data distribution depends on the current policy
   → Policy changes → Data changes → Moving target
4. **No gradient through the environment:** reward is a black box
   → Requires gradient estimation



Explore-Exploit dilemma in Multi-armed bandit problem, spend all the money on one slot machine or share it between other machines?

Wikipedia Commons - Multi-Armed Bandit problem CC BY-SA 3.0

2.4 Foundation - Why Molecules?

Optimization across a vast search space using sequential, goal-driven decision-making without ground-truth labels

1. **Vast search space:** up to 10^{60} drug-like molecules (*Bohacek et al., Med. Res. Rev., 1996*)

   → Enumeration infeasible

2. **Black-box objective:** ML models, docking, synthesizability

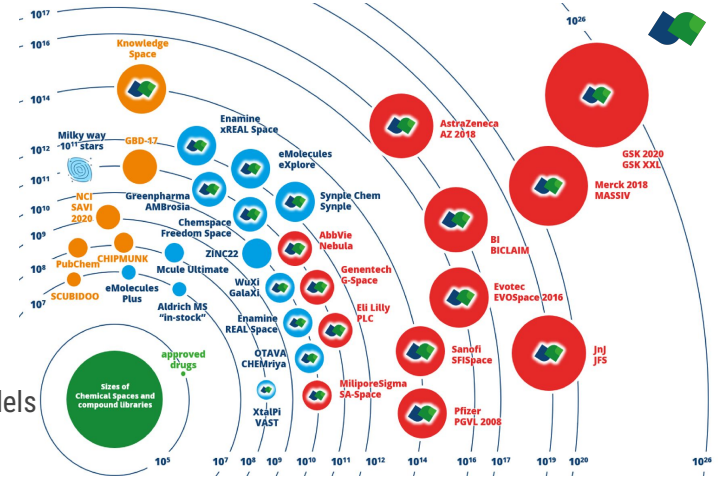
   → Non-differentiable w.r.t. molecular structure

3. **No "correct molecule" labels**: imitation of the training distribution by generative models

   → Goal: optimize, not imitate

4. **Sequential construction**: molecules are built atom by atom or fragment by fragment

   → Natural sequential decision process



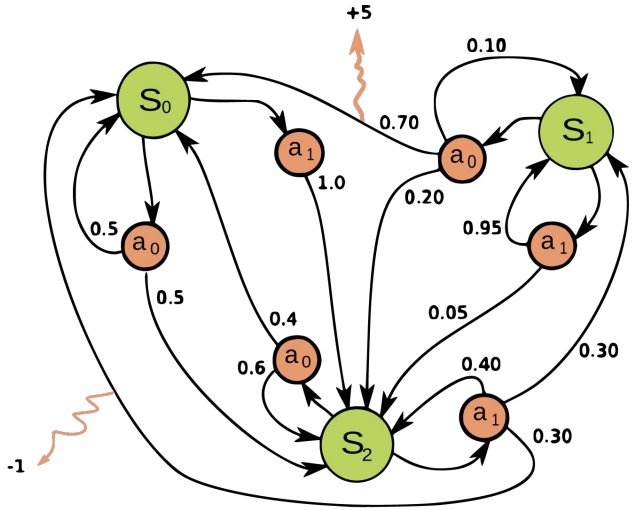
Magnitude of search spaces for early-phase drug discovery

2.5 Foundation - Markov Decision Processes (MDPs)

The mathematical universe of decision-making: states, stochastic transitions, expected rewards, discount factors, and the memoryless Markov property

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$$

- $\mathcal{S}$: state space
- $\mathcal{A}$: action space
- P : transition dynamics
- R : reward function
- $\gamma \in [0, 1]$: discount factor



$$P(s' \mid s, a) = \Pr\left(S_{t+1} = s' \mid S_t = s,\ A_t = a\right)$$

$$R(s, a) = \mathbb{E}\left[R_{t+1} \mid S_t = s,\ A_t = a\right]$$

Markov Property: the present state contains all relevant history

$$\Pr(S_{t+1} \mid S_t, A_t, S_{t-1}, A_{t-1}, \ldots, S_0) = \Pr(S_{t+1} \mid S_t, A_t)$$

Wikimedia Commons, "Markov Decision Process.svg", CC BY-SA 4.0.

2.6 Foundation - Trajectories & Return

The reward of future actions affecting today's reward

- **Trajectory**: episodic

$$\tau = (S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_T)$$

- **Return** (or **Gain**):

  How much gain brings each action?

  Gain belongs to step of a trajectory

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = \sum_{k=0}^{T-t-1} \gamma^k R_{t+k+1}$$

$$G_t = R_{t+1} + \gamma G_{t+1}$$

$\gamma \rightarrow$ 0: **myopic**

$\gamma \rightarrow$ 1: **far-sighted**

- Models the reality: future is uncertain!
- An action today is worth more than tomorrow!
- Keeps the infinite sum finite!

Sutton & Barto, Reinforcement Learning: An Introduction, 2nd ed., MIT Press, 2018

2.7 Foundation - The Policy: π(a|s)

Transforming stochastic state-to-action distributions into trajectory probabilities to unlock policy gradient optimization

- **Parameterized policy (softmax over logits):**

$$\pi_\theta(a \mid s) = \frac{\exp\left(f_\theta(s,a)\right)}{\sum_{a' \in \mathcal{A}(s)} \exp\left(f_\theta(s,a')\right)}$$

A(s): *State-dependant valid action*

f_θ: Any network (MLP, RNN,Transformer) outputs logits

- **Trajectory probability:** the holy grail for policy gradient methods such as REINFORCE & PPO

$$\tau = (S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_T)$$

$$p_\theta(\tau) = p(S_0) \prod_{t=0}^{T-1} \pi_\theta(A_t \mid S_t)\, P(S_{t+1} \mid S_t, A_t)$$

Sutton & Barto, Reinforcement Learning: An Introduction, 2nd ed., MIT Press, 2018

2.8 Foundation - Value Functions & Advantage

Evaluation of the World: reinforcing the good and suppressing the bad

- **State Value:** how good is the state under π

$$V^{\pi}(s) = \mathbb{E}_{\pi}\left[G_t \mid S_t = s\right]$$

- **Action (Q) Value:** how good is this action in this state, then following π

$$Q^{\pi}(s,a) = \mathbb{E}_{\pi}\left[G_t \mid S_t = s,\ A_t = a\right]$$

where $G_t = R_{t+1} + \gamma G_{t+1}$

- **Advantage function:** how much better than average this action is

$$A^{\pi}(s,a) = Q^{\pi}(s,a) - V^{\pi}(s)$$

$$\mathbb{E}_{a\sim\pi}\left[A^{\pi}(s,a)\right] = 0$$

- **$A > 0$: reinforce**
- **$A < 0$: suppress**

Sutton & Barto, Reinforcement Learning: An Introduction, 2nd ed., MIT Press, 2018,

2.9 Foundation - Molecule Design as MDP

Translating chemical generation into an MDP framework featuring deterministic structural dynamics and terminal-only sparse scoring

MDP Element	Molecule Design
State S_t	Partial molecule (SMILES prefix / graph / fragment assembly)
Action A_t	Append token / add SMILES character / attach next fragment
Transition *P*	Deterministic, known
Reward	Sparse, terminal-only
γ	1(short, episodic)
Termination	Fixed length or stop action

$$R_{t+1} = 0 \;\; (t < T-1), R_T = s(m_T)$$

γ = 1, then

$$G_t = R_T = s(m_T) \quad \forall\, t$$

$$p_\theta(\tau) = p(S_0) \prod_{t=0}^{T-1} \pi_\theta(A_t \mid S_t)\, P(S_{t+1} \mid S_t, A_t)$$

Dynamic of the world is deterministic!

$$p_\theta(\tau) = \prod_{t=0}^{T-1} \pi_\theta(A_t \mid S_t) \;\Rightarrow\; \log p_\theta(m) = \sum_{t=0}^{T-1} \log \pi_\theta(A_t \mid S_t)$$

2.10 Foundation - Taxonomy Map

Navigating the RL taxonomy to isolate and master model-free, policy-based optimization variants

Value-based:

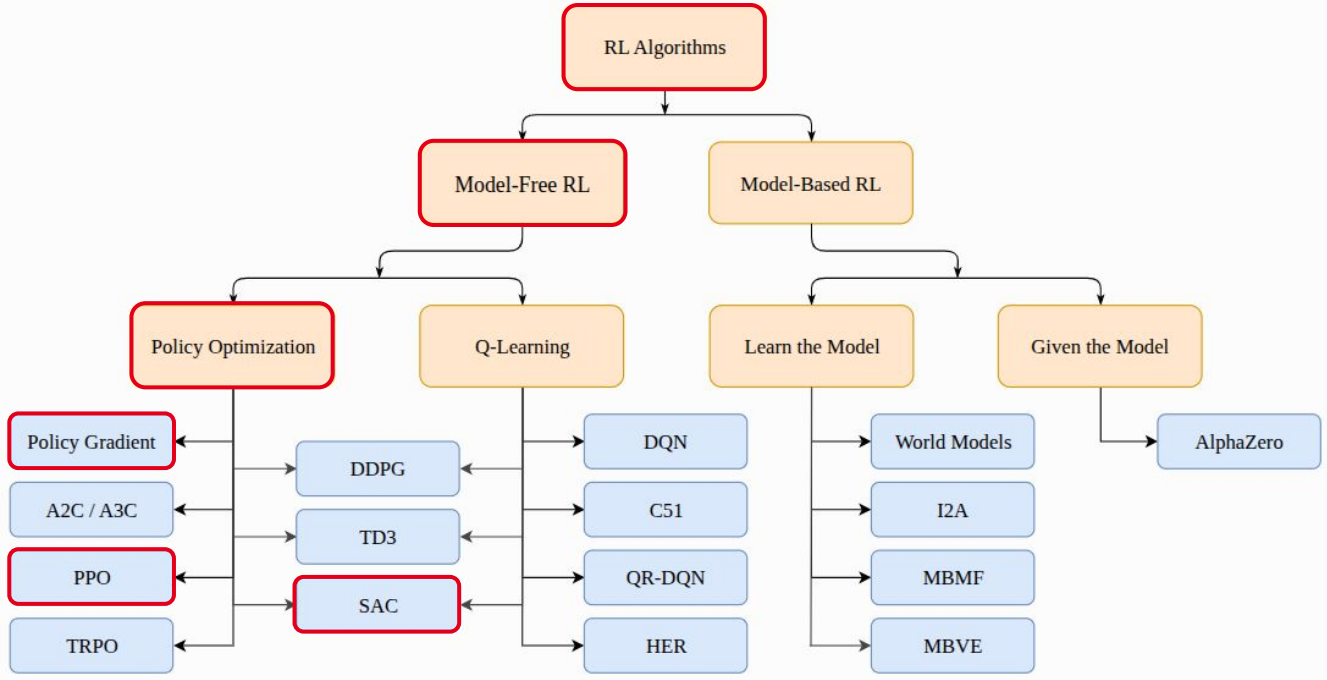
$$\pi(s) = \arg\max_a Q_\phi(s, a)$$

Policy-based:

$$\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$$

Actor-critic:

$$\pi_\theta(a \mid s) \; + \; V_\phi(s)$$



J. Achiam, "Spinning Up in Deep RL", OpenAI, 2018

2.11 Foundation - The Objective

Maximizing expected return over a distribution we control, but cannot differentiate through

$$J(\theta) = \mathbb{E}_{\tau \sim p_\theta}\big[G(\tau)\big], \qquad G(\tau) := G_0 = \sum_{t=0}^{T-1} \gamma^t R_{t+1}$$

In context of molecule design: $J(\theta) = \mathbb{E}_{m \sim p_\theta}\big[s(m)\big]$

Ultimate goal:

$$\theta^* = \arg\max_\theta J(\theta)$$

$$\nabla_\theta J(\theta) = \sum_\tau \nabla_\theta p_\theta(\tau)\, G(\tau) \quad \leftarrow \text{not an expectation!}$$

Brick wall

1. **θ** lives in the sampling distribution, not in G
2. G is a black box: no ∇ through the reward
3. **∇p_θ** is not a probability, so there are no samples to average

   →i.e. no Monte Carlo estimation
4. **Reparameterization trick** (VAEs) does not apply

2.12 Foundation - The Log-Derivative Trick

Turning the gradient of an expectation into an expectation of a gradient

$$\nabla_\theta p_\theta(\tau) = p_\theta(\tau)\,\nabla_\theta \log p_\theta(\tau)$$

Likelihood Ratio Trick

$$\frac{d}{dx}\log f(x) = \frac{1}{f(x)}\frac{df}{dx}$$

Derivation:

$$\begin{aligned}
\nabla_\theta J(\theta) &= \nabla_\theta \sum_\tau p_\theta(\tau)\,G(\tau) \\
&= \sum_\tau \nabla_\theta p_\theta(\tau)\,G(\tau) \\
&= \sum_\tau p_\theta(\tau)\,\nabla_\theta \log p_\theta(\tau)\,G(\tau) \\
&= \mathbb{E}_{\tau\sim p_\theta}\Big[G(\tau)\,\nabla_\theta \log p_\theta(\tau)\Big]
\end{aligned}$$

- Also called the **score-function** or **likelihood-ratio estimator**
- Requirements: **G** independent of **θ**; **p_θ** differentiable in **θ**
- Result: an expectation, so it can be estimated from samples
- **G** needs no gradient: a black-box oracle is fine

That's why docking scores and ML models work as rewards

R. J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. Machine Learning 8, 229–256, 1992.

2.13 Foundation - Policy Gradient Theorem

The environment dynamics drop out, so learning requires no model of the world

Log of the trajectory probability: The log turns the product into a sum, and P(s'|s,a) has no θ, so its gradient is zero

$$\log p_\theta(\tau) = \underbrace{\log p(S_0)}_{=0} + \sum_{t=0}^{T-1} \log \pi_\theta(A_t \mid S_t) + \underbrace{\sum_{t=0}^{T-1} \log P(S_{t+1} \mid S_t, A_t)}_{=0}$$

Only the policy term depends on θ:

$$\nabla_\theta \log p_\theta(\tau) = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(A_t \mid S_t)$$

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta}\Big[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(A_t \mid S_t)\, G(\tau)\Big]$$

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta}\Big[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(A_t \mid S_t)\, G_t\Big]$$

- Model-free: we only need to *sample* the environment, never model it
- G_t instead of $G(\tau)$: same expectation, lower variance

$$\nabla_\theta J(\theta) \propto \mathbb{E}_{s \sim d^\pi,\, a \sim \pi_\theta}\Big[Q^\pi(s,a)\, \nabla_\theta \log \pi_\theta(a \mid s)\Big]$$

d^π: the state-visitation distribution under π

R. J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. Machine Learning 8, 229–256, 1992.

2.14 Foundation - REINFORCE

Monte Carlo policy gradient: sample, score, reweight

Monte Carlo gradient estimate from N sampled trajectories:

$$\hat{g} = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T-1} G_t^{(i)} \nabla_\theta \log \pi_\theta \left(A_t^{(i)} \mid S_t^{(i)} \right)$$

The weight update via standard gradient ascent:

$$\theta \leftarrow \theta + \alpha \hat{g}$$

Surrogate loss for autograd (G treated as a constant):

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T-1} \mathrm{sg}\left[G_t^{(i)}\right] \log \pi_\theta \left(A_t^{(i)} \mid S_t^{(i)} \right)$$

repeat:

1. Sample N trajectories with π_θ
2. Compute returns G_t for each step
3. L = −mean(G_t · log π_θ(A_t | S_t))
4. θ ← θ − α ∇L

- Interpretation: return-weighted maximum likelihood on the agent's own samples
- Unbiased, simple, critic-free, but high variance

R. J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. Machine Learning 8, 229–256, 1992.

2.15 Foundation - Variance & Baseline

Subtracting a reference value: zero bias, far less noise

Policy gradient with a baseline:

$$\nabla_\theta J(\theta) = \mathbb{E}\Big[\sum_t \nabla_\theta \log \pi_\theta(A_t \mid S_t)\,\big(G_t - b(S_t)\big)\Big]$$

The proof of unbiasedness:

$$\mathbb{E}_{a\sim\pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, b(s)\big] = b(s)\sum_a \pi_\theta(a \mid s)\,\nabla_\theta \log \pi_\theta(a \mid s)$$

$$= b(s)\sum_a \nabla_\theta \pi_\theta(a \mid s) = b(s)\,\nabla_\theta \sum_a \pi_\theta(a \mid s) = b(s)\,\nabla_\theta 1 = 0$$

$$b(s) = V^\pi(s) \;\Rightarrow\; G_t - V^\pi(S_t) \approx A^\pi(S_t, A_t)$$

- Problem: all rewards positive → every sampled action is pushed up; only the relative size of the push carries signal
- **Baseline**: centers the weights, so better-than-expected → up and worse → down
- Constraint: ***b*** may depend on the **state**, never on the action ***A_t***
- Best common choice: $b \approx V^\pi$, and the weight becomes the advantage

2.16 Foundation - Choosing the Baseline

From critics to RLOO: estimating “average” without learning a value function

- **Option 1**: leave-one-out over *k* samples for the same context x

$$\tau_1, \dots, \tau_k \sim \pi_\theta(\cdot \mid x), \qquad r_i = G(\tau_i)$$

$$b_i = \frac{1}{k-1} \sum_{j \neq i} r_j$$ Leave one out!

$$\hat{g}_{\mathrm{RLOO}} = \frac{1}{k} \sum_{i=1}^{k} \left(r_i - b_i\right) \nabla_\theta \log \pi_\theta(\tau_i \mid x)$$ Estimate the gradient!

$$r_i - b_i = \frac{k}{k-1}\left(r_i - \bar{r}\right), \qquad \bar{r} = \frac{1}{k} \sum_{j=1}^{k} r_j$$

- **Option 2**: learned critic (actor-critic)

$$b(S_t) = V_\phi(S_t), \qquad \mathcal{L}(\phi) = \mathbb{E}\left[(G_t - V_\phi(S_t))^2\right]$$

Kool, van Hoof & Welling, ICLR Workshop 2019; Ahmadian et al., ACL 2024; Shao et al., arXiv:2402.03300, 2024

2.17 Foundation - Actor-Critic (AC) & Policy Proximal Optimization (PPO)

Reusing data safely: trust regions through clipping

TD-based advantage (actor-critic):

$$\delta_t = R_{t+1} + \gamma V_\phi(S_{t+1}) - V_\phi(S_t) \qquad V(s) = \mathbb{E}[\text{future return} \mid s]$$

Generalized advantage estimation (GAE):

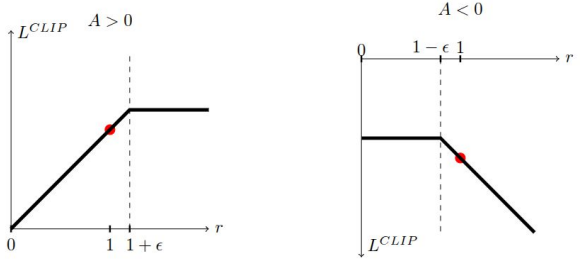
$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \, \delta_{t+l}$$

Probability ratio (importance sampling):

$$\rho_t(\theta) = \frac{\pi_\theta(A_t \mid S_t)}{\pi_{\theta_{\text{old}}}(A_t \mid S_t)}$$

Clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(\rho_t(\theta)\,\hat{A}_t,\ \text{clip}\Big(\rho_t(\theta), 1-\epsilon, 1+\epsilon\Big)\,\hat{A}_t\Big)\Big]$$



Full PPO objective (maximization):

$$L(\theta, \phi) = L^{\text{CLIP}}(\theta) - c_1\,\mathbb{E}_t\left[(V_\phi(S_t) - G_t)^2\right] + c_2\,\mathbb{E}_t\left[\mathcal{H}[\pi_\theta(\cdot \mid S_t)]\right]$$

Schulman et al., ICML 2015 (TRPO); Schulman et al., ICLR 2016 (GAE); Schulman et al., 2017 (PPO)

2.18 Foundation - The Red Thread

Sample → Score → Compare → Shift probability

Agent

Environment

Policy

Reward

What do we want?

High reward, on average

$$J(\theta) = \mathbb{E}[G]$$

Why it is hard?

The reward is a black box

$$\nabla_\theta s(m) = ?$$

How do we get a gradient anyway?

Differentiate the probability, not the reward

$$\nabla \log \pi_\theta$$

Do we need to know the world?

No: the physics cancels out

$$\nabla \log P = 0$$

What does learning do?

Good outcomes become more likely

$$G \cdot \nabla \log \pi_\theta$$

How do we make it stable?

Compare to average, not to zero

$$G - b$$

RLOO: the other samples are the average

PPO: do not move too far at once

3 Landscape - Literatures by Taxonomy

1

SOFTWARE Open Access

Reinvent 4: Modern AI–driven generative molecule design

Hannes H. Loeffler[1*], Jiazhen He[1], Alessandro Tibo[1], Jon Paul Janet[1], Alexey Voronov[1], Lewis H. Mervin[2] and Ola Engkvist[1]

Journal of Cheminformatics 2024

2

OPEN

Bidirectional reinforcement learning neural network for constrained molecular design

Junan Lin[1], Jiří Hostaš[1], Anguang Hu[2], Hang Hu[1], Hsu Kiang Ooi[1] & Mohammad Sajjad Ghaemi[1]

Scientific Reports 2025

3

SHARP: Generating Synthesizable Molecules via Fragment-Based Hierarchical Action-Space Reinforcement Learning for Pareto Optimization

Jeonghyeon Kim, Seongok Ryu, Woohyeong Lee, Ansoo Lee, Hahnbeom Park,* and Chaok Seok*

Journal of Chemical Information and Modeling 2025

3.1 Landscape - REINVENT 4

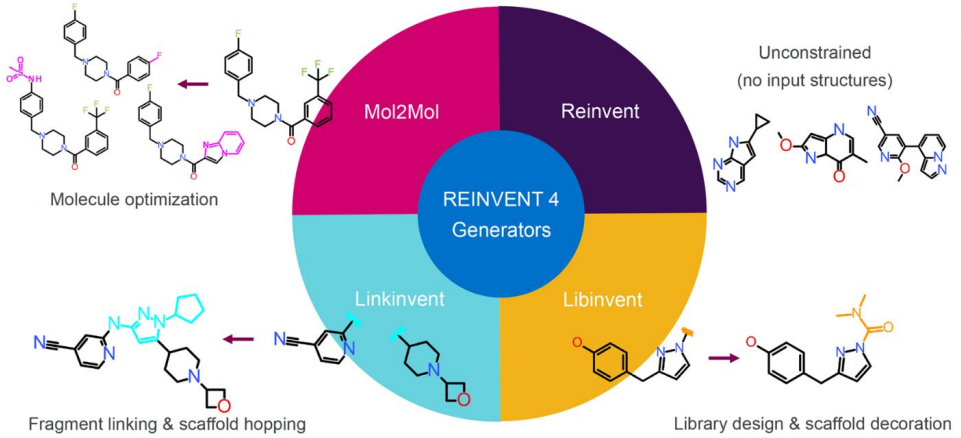
Flexible, modular framework for generative molecular design

- **Metadata:**
  Journal of Cheminformatics (2024) - Astrazeneca

- **Limitation:**
  Early molecular RL tools were locked
  into **single** generative models (SMILES RNNs)

- **Core Objective:**
  - An all-in-one framework supports different drug discovery tasks
  - Multi-objective RL optimization towards user-defined properties while retaining chemical validity

- **Architecture:**
  - **Priors:** fixed pretrained foundation models on ChEMBL data
  - **Agent:** biases the probability distribution of generated molecules towards candidate with desired properties



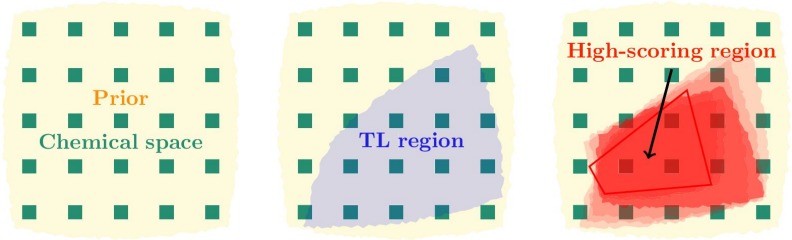
Loeffler et al., J Cheminform (2024)

3.2 Landscape - REINVENT 4

Sequential RL runs tune the performance of foundation models for multi-parameter optimization

- RL Mechanics (MDP Formulation):
  - **State:** partial sequence or molecular representation assembled step-by-step
  - **Action:** selecting the next token/fragment in the sequence generator
  - **Reward:** multi-objective scoring function combining **QSAR** predictive models, **structural** alerts and **physicochemical** properties

- Technical Rationale:
  - **Policy Gradient via Augmented Likelihood and DAP Strategy:**
    - Augmented Likelihood instead of PPO/REINFORCE : $\log P_{\text{aug}}(T) = \log P_{\text{prior}}(T) + \sigma S(T)$
    - Difference between Augmented and Posterior (DAP) : $\mathcal{L}(T) = [\log P_{\text{aug}}(T) - \log P_{\text{agent}}(T)]^2$
  - **Transfer Learning:** retraining priors using teacher-forcing strategy
  - **Staged Learning:** different scoring profiles can be introduced sequentially

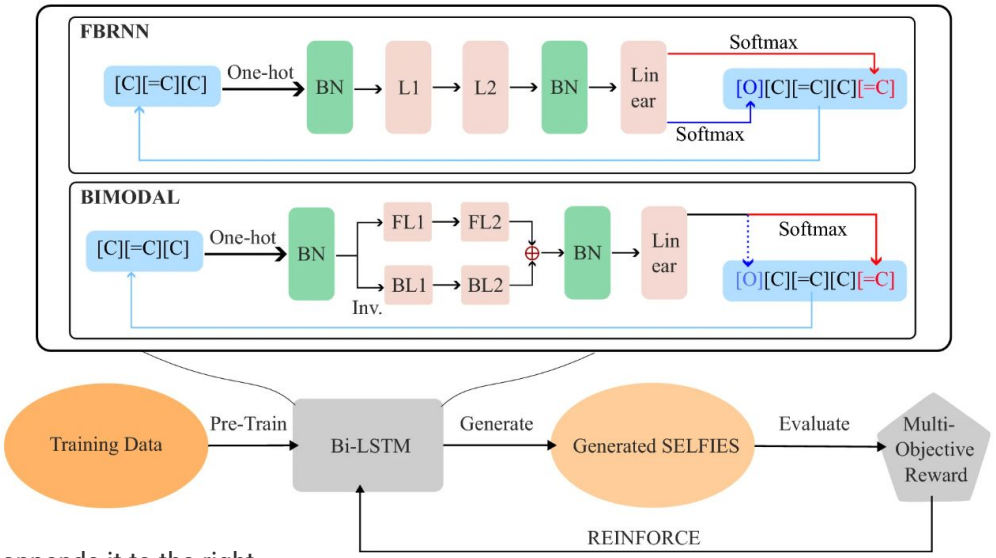


Loeffler et al., J Cheminform (2024)

3.3 Landscape - BiRLNN

Simultaneous bidirectional generation of SMILES or SELFIES controlled by RL

- Metadata:
  Scientific Reports (2025)
- Focus Area:
  Constrained molecular generation with desired property
- Limitation:
  Unidirectional autoregressive generators suffer restricted chemical space coverage



- Architecture: comparison of 3 distinct generation paradigms
  - **Forward or Backward RNN:** predicts the next token and appends it to the right
  - **Forward and Backward RNN (FB-RNN):** predicts two tokens per iteration: one for each end
  - **BIMODAL (once Forward, once Backward):** uses information from both directions to predict the next token in an alternating generation process

```
torch.nn.LSTM(bidirectional = True)
```

3.4 Landscape - BiRLNN

RL increases the probability of generating molecules with desirable properties

- RL Mechanics - The Fine-Tuning Loop :
  1. **Pretrained Bi-LSTM:**

     Start with a model that has already learned molecular sequence patterns.
  2. **Generate a batch of molecules**

     A trajectory: policy samples complete molecular strings.
  3. **Calculate rewards**

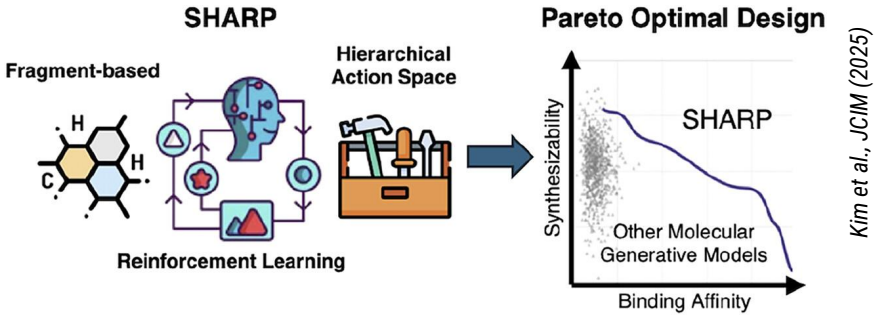
     Evaluate the completed molecules: **QED** and **SA**, combination of the scores into **one scalar** reward
  4. **Calculate the policy gradient**

     Use **REINFORCE** to increase the likelihood of trajectories with **batch-moving-average** rewards: $A(s_t) = R(s_T) - b(s_t)$
  5. **Update the LSTM parameters**

3.5 Landscape - SHARP

Structure-based fragment growing with multi-objective optimization

- Metadata:
  Journal of Chemical Information and Modeling (2025)
- Focus Area:
  3D structure-based molecule generation



Kim et al., JCIM (2025)

- Limitations:
  - Atom-by-atom generation in **3D pocket** space easily violates chemical **synthesizability**
  - **2D** SMILES methods ignore **3D** pocket shape
  - Single scalar multi-objective rewards fail to capture trade-offs

- Architecture:
  - **Synthesizability Estimation Model (SEM) :** masks actions; special atoms, bonds, fragments are allowed
  - **RL Agent:**
    - **Main Policy**: chooses an operation; add, delete or substitute
    - **Subpolicy**: selects fragments for chosen operation
    - **Critic Network**: estimates the value of current state

3.6 Landscape - SHARP

Pareto-analysis of generated molecules for assessment the trade-off among multi-objectives

- RL Mechanics (MDP Formulation):
  - **State:** 3D complex of the protein pocket and the currently grown 3D molecular fragment structure
  - **Hierarchical Action-Space:**
    - ***High-Level :*** Select types of modification of perform
    - ***Low-Level :***
      - If add, then select atom or fragment to add and attachment point
      - If delete, then select a bond to break
      - if substitute, then select a bond to break, fragment to replace and the attachment config
  - **Reward Vector:** 45% docking score, pharmacophore similarity 45% and 10% SASA **(*)**

- Algorithm Choice:
  - **Soft Actor-Critic (SAC) :**
    - Off-policy; learning from previous experiences
    - Critic estimates the value of state-action pair : $Q(s,a) = \mathbb{E}[\text{future return} \mid s,a]$
  - **Pareto Optimization via reward vector (*)**

3.7 Landscape - Conclusion

Three different approaches to RL-based molecular design

- REINVENT 4
  - Prior-anchored likelihood optimization
  - Changes how the generative model is optimized
  - The agent is guided toward high-scoring molecules while remaining anchored to a pretrained prior

- BiRLNN
  - Bidirectional sequence generation
  - Changes how the molecule is generated
  - The model grows sequences from both ends and uses REINFORCE to optimize molecular properties

- SHARP
  - Hierarchical fragment-based RL
  - Changes the molecular action space
  - The agent selects molecular operations and fragments, with synthesizability constrain

4.1 Case Study - Inspiration

Proposal of a policy-based approach for famous combinatorial optimization problem: Traveling Salesman

Published as a conference paper at ICLR 2019

ATTENTION, LEARN TO SOLVE ROUTING PROBLEMS!

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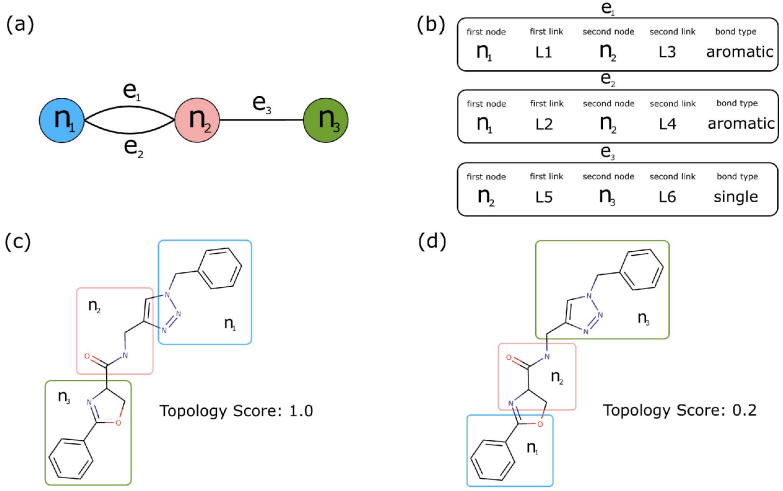
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ABSTRACT

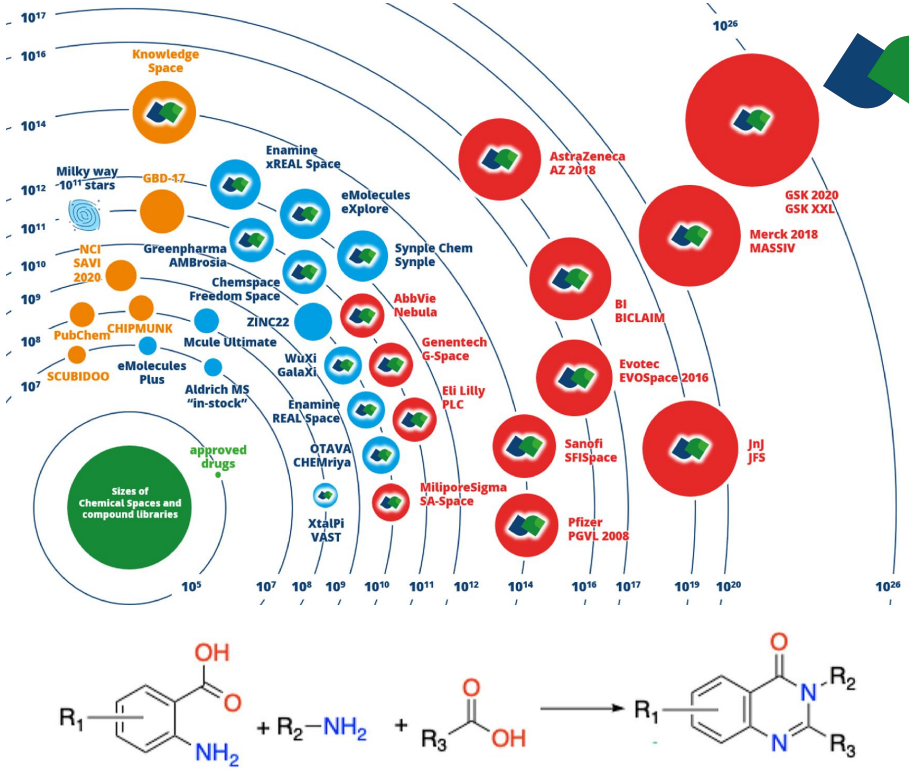
The recently presented idea to learn heuristics for combinatorial optimization problems is promising as it can save costly development. However, to push this idea towards practical implementation, we need better models and better ways of training. We contribute in both directions: we propose a model based on attention layers with benefits over the Pointer Network and we show how to train this model using REINFORCE with a simple baseline based on a deterministic greedy rollout, which we find is more efficient than using a value function. We significantly improve over recent learned heuristics for the Travelling Salesman Problem (TSP), getting close to optimal results for problems up to 100 nodes. With the same hyperparameters, we learn strong heuristics for two variants of the Vehicle Routing Problem (VRP), the Orienteering Problem (OP) and (a stochastic variant of) the Prize Collecting TSP (PCTSP), outperforming a wide range of baselines and getting results close to highly optimized and specialized algorithms.

4.2 Case Study - Combinatorial Explosion

Exponential growth of the make on demand molecules



Bellmann et al., *J. Chem. Inf. Model., 2021*



Combinatorial Explosion: 376K **x** 500K **x** 500K **~** **9.4 x 10¹⁵ !!**

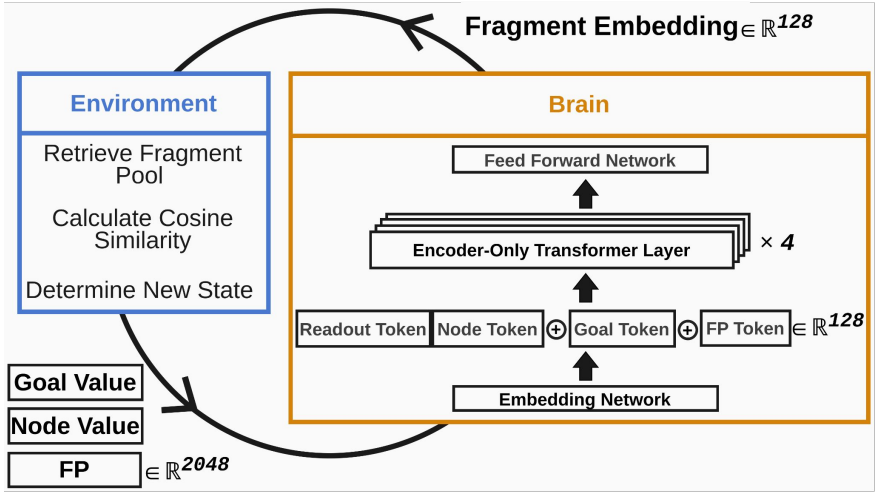
Klarich et al., *J. Chem. Inf. Model., 2024*

4.3 Case Study - Goal-Conditioned RL over Combinatorial Fragment Spaces

Attentive-based policy learns goal-directed combination of fragments directly from topological fragment spaces

- Focus Area:
  - Extraction of molecules with desired property directly from chemical fragment space
  - Over 80% chance of synthesizability within few weeks

- Core Objective:
  - Optimization of ML model's prediction toward desired outputs
  - Training a goal-conditioned policy for fragment combination
  - **Amortization** over any goal value at inference time

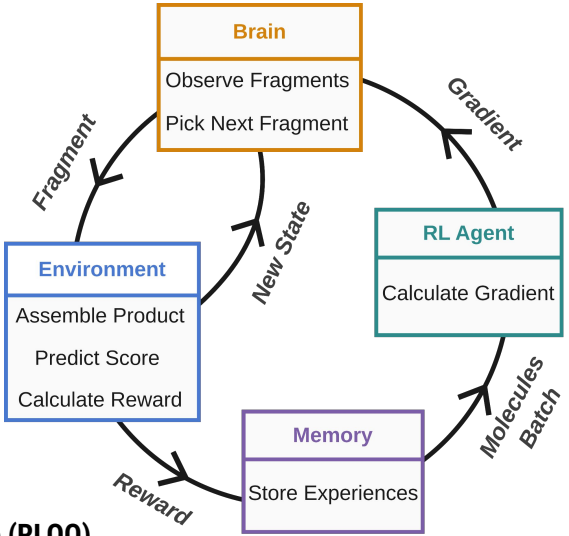


- Architecture:
  - **Embedder Layers:** goal value, fragments active bit,
  - **Attention Layer:** encoder-only transformer to capture interaction between fragments based on goal presents in given state
  - **Linear Layers:** converts attentive representation of so far selected fragments back to **128D** space

4.4 Case Study - Goal-Conditioned RL over

Optimizing the property prediction of compounds extracted from make on demand spaces via RLOO

- RL Mechanics (MDP Formulation):
  - **State**:
    - partial molecules (fragments sofar)
    - current node of topology
    - goal value of specific property
  - **Action**: select a fragment from fragments pool of current node
  - **Reward**: sparse, terminal-only reward
  - **Score Function**: surrogate ML model responsible for prediction of molecular property (logP, ADME profiles, affinity, …)



- Algorithm Choice: **REINFORCE with with Leave-One-Out Baseline (RLOO)**
  - **Short horizon :** 3 steps, 3 fragments to pick
  - **Critic or baseline:** simply mean of batch
  - **Oracle calls:** cheap ML model at inference time
  - **Entropy:** within group diversity
  - **batch size = 16:** variance within the batch stays high

5 References

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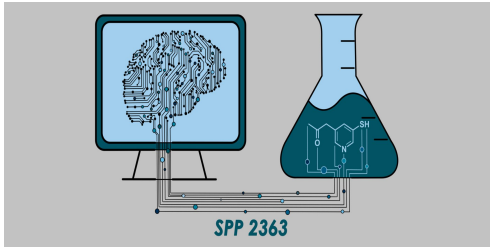


Current (alphabetic order): Christiane Ehrt, Gerd Embruch, Joel Graef, Melanie Geringhoff, Malte Holmer, Paula Kammer, Tim Kuhrt, Eileen Kränzle, Nele klatte, Christoph Noak, Maarten Mäcking, Marius Rüve, Malte Schokolowski, Thorben Schulze & Hadi Vareno

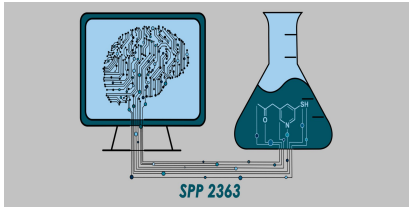
Many Thanks for Your Attention

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Key Takeaways



1. The Problem: What Are We Trying to Solve?

Navigating enormous chemical spaces to discover promising molecular candidates under multiple objectives and constraints

2. The Motivation: Why Reinforcement Learning?

Understanding when sequential, experience-driven decision-making offers advantages over Genetic Algorithms, Bayesian Optimization and Thompson Sampling

3. The Architecture: Components of an RL System

Agent, environment, states, actions, rewards and policy. And how they interact to generate experience

4. The Learning Mechanism: How Does RL Optimize?

From policy gradients and likelihood ratios to variance reduction, advantage estimation and stable policy updates



Supplementals

Foundation - Value-Based Learning

Bellman equations

Bellman expectation equation :

$$V^{\pi}(s) = \sum_{a} \pi(a \mid s) \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V^{\pi}(s') \right]$$

$$Q^{\pi}(s,a) = R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \sum_{a'} \pi(a' \mid s')\, Q^{\pi}(s',a')$$

Bellman optimality equation :

$$V^{*}(s) = \max_{a} \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V^{*}(s') \right]$$

$$Q^{*}(s,a) = R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, \max_{a'} Q^{*}(s',a')$$

Optimal policy: $\pi^{*}(s) = \arg\max_{a} Q^{*}(s,a)$

Foundation - Value-Based Learning

From tabular to deep neural networks

Tabular Q-learning update:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \Big[\underbrace{R_{t+1} + \gamma \max_{a'} Q(S_{t+1}, a')}_{\text{TD target } y_t} - Q(S_t, A_t) \Big]$$

Temporal Difference (TD) error:

$$\delta_t = R_{t+1} + \gamma \max_{a'} Q(S_{t+1}, a') - Q(S_t, A_t)$$

Deep-Q-Network (DQN) loss:

$$\mathcal{L}(\phi) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[\left(r + \gamma \max_{a'} Q_{\phi^-}(s', a') - Q_{\phi}(s, a) \right)^2 \right]$$

Foundation - Limits of Value-Based Methods

Limits of value-based methods

Huge or variable action spaces: argmax requires enumerating every valid action

→ Fixed output heads break when $\mathcal{A}(s)$ changes size from state to state:

$$\pi(s) = \arg\max_{a \in \mathcal{A}(s)} Q_\phi(s,a) \quad \Rightarrow \quad \mathcal{O}\Big(\big|\mathcal{A}(s)\big|\Big) \text{ evaluations per step}$$

Continuous actions: argmax becomes an optimization problem of its own

Deterministic greedy policy: no inherent diversity; ε-greedy exploration is undirected

Discontinuous policy: a small change in Q can flip the argmax → unstable behavior

Overestimation bias: max over noisy estimates, systematic upward bias (Double DQN, *van Hasselt et al., AAAI 2016)*

$$\mathbb{E}\Big[\max_a \hat{Q}(s,a)\Big] \;\geq\; \max_a \mathbb{E}\Big[\hat{Q}(s,a)\Big]$$